EXPERIMENT NO. 4 DEPLOYMENT AND NETWORK CONFIGURATION OF KALI LINUX AND VULNERABLE VMS ON AN ISOLATED VIRTUAL NETWORK

AIM:

To deploy Kali Linux and vulnerable virtual machines, configure an isolated virtual network, and establish controlled communication between the systems for cybersecurity testing and analysis.

OBJECTIVES:

1. To understand the deployment and configuration of Kali Linux in a virtualized environment.
2. To deploy vulnerable virtual machines for controlled cybersecurity testing.
3. To configure an isolated virtual network for communication between virtual machines.
4. To verify IP addressing and network connectivity between the virtual systems.
5. To understand the importance of network isolation during security testing.

HARDWARE & SOFTWARE REQUIREMENTS:

Category	Specification & Tools
Hardware Configuration	PC/Laptop with Intel Core i5 or higher, minimum 4 GB RAM, Internet Connection
Operating System	Windows 10/11
Network	Host-only / Isolated Virtual Network
Virtualization Platform	VMware Workstation / VirtualBox
Attacker VM	Kali Linux
Vulnerable VM	DVWA / Metasploitable / Vulnerable Linux VM
Tools	Kali Linux networking and security tools
Documentation Tools	Microsoft Word

THEORY:

Virtualization is a technology that allows multiple operating systems to run as separate virtual machines (VMs) on a single physical computer. It is commonly used in cybersecurity laboratories because different systems can be deployed and tested in a controlled environment without affecting the main operating system.

Kali Linux is a Linux-based operating system designed for cybersecurity and security testing. It provides various tools for network analysis, vulnerability assessment, penetration testing, digital forensics, and security auditing. In this practical, Kali Linux is used as the security testing machine.

A vulnerable virtual machine is a system intentionally configured with known security weaknesses. Such machines are useful for learning and practicing security testing in a controlled laboratory environment without targeting real systems.

An isolated virtual network is a private network created within the virtualization platform that allows selected virtual machines to communicate with each other while limiting their access to the external network. This isolation helps prevent vulnerable systems from being exposed to the Internet or other production devices.

Network configuration involves assigning and checking IP addresses, subnet masks, gateways, and network interfaces of the virtual machines. Proper configuration ensures that Kali Linux and the vulnerable VM can communicate over the intended virtual network.

Connectivity verification is performed after configuration to confirm that the virtual machines can reach each other. Commands such as ip addr can be used to view network settings, while ping can be used to test basic communication between the systems.

The combination of Kali Linux, vulnerable VMs, and an isolated virtual network provides a controlled environment for studying cybersecurity concepts, testing security tools, and analyzing vulnerabilities without unnecessarily affecting external systems.

VIRTUAL MACHINE DEPLOYMENT & NETWORK CONFIGURATION – PHASES, TOOLS & COMMANDS:

1. **Kali Linux Deployment**
   In this phase, Kali Linux is deployed as a virtual machine using a virtualization platform. The VM is configured with the required CPU, memory, storage, and network settings before starting the system.

Tool / Utility	Purpose	Command Used
Virtualization Platform	Deploy Kali Linux VM	Import/Start VM
Kali Linux	Security testing system	Start VM

Tool / Utility	Purpose	Command Used
Kali Terminal	Identify attacker IP	ifconfig

2. Vulnerable VM Deployment

In this phase, a vulnerable virtual machine is deployed to provide a controlled target for cybersecurity testing. The VM is configured separately and connected to the intended isolated virtual network.

Tool / Utility	Purpose	Command Used
Virtualization Platform	Create/import vulnerable VM	Create/Import VM
Vulnerable VM	Controlled target system	Start VM
VM Settings	Configure system resources	CPU, RAM, Storage

3. Isolated Virtual Network Configuration

In this phase, Kali Linux and the vulnerable VM are connected to the same isolated virtual network. The network configuration prevents the vulnerable system from being directly exposed to the external network.

Tool / Utility	Purpose	Command / Action Used
Virtualization Platform	Create isolated network	Host-Only
Virtual Network	Provide private communication	192.168.56.0/24
Kali Linux	Verify network interface	ifconfig

4. IP Address Configuration

In this phase, the IP addresses and network interfaces of both virtual machines are checked. Proper IP configuration ensures that Kali Linux and the vulnerable VM are placed on the same subnet and can communicate with each other.

Tool / Utility	Purpose	Command / Action Used
Kali Terminal	Identify Kali IP	ifconfig

Tool / Utility	Purpose	Command / Action Used
Metasploit	Set target IP	set RHOSTS 192.168.56.103
Metasploit	Set local IP	set LHOST 192.168.56.102
Metasploit	Verify configuration	show options

5. **Network Connectivity Verification**
   In this phase, connectivity between Kali Linux and the vulnerable VM is verified using the ping command. Successful responses confirm that both systems can communicate over the isolated virtual network.

Tool / Utility	Purpose	Command / Action Used
Nmap	Scan target services and OS	nmap -A 192.168.56.103
Nikto	Scan HTTP service	nikto -h http://192.168.56.103
Metasploit	Launch framework	msfconsole
Metasploit	Search relevant module	search vsftpd
Metasploit	Select module	use 1
Metasploit	Verify module settings	show options

IMPLEMENTATION STEPS:

1. **Kali Linux Deployment**
   - Opened the virtualization platform and imported the Kali Linux virtual machine.
   - Configured the required virtual hardware resources and started Kali Linux.
   - Connected Kali Linux to the Host-Only isolated network.
   - Opened the Kali terminal and used ifconfig to identify the attacker's IP address.
   - Verified the Kali Linux system was ready for security testing.

2. **Vulnerable VM Deployment**
   - Imported and started the Metasploitable2 vulnerable virtual machine.
   - Configured the required virtual hardware and connected it to the same isolated network.
   - Verified that the vulnerable VM was running correctly.
   - Identified the target IP address as 192.168.56.103 from the provided setup.

- Confirmed that the vulnerable VM was available as the controlled target for security testing.

3. Isolated Virtual Network Configuration

- Configured a Host-Only virtual network using the 192.168.56.0/24 address range.
- Connected both Kali Linux and Metasploitable2 to the same isolated virtual network.
- Verified that the attacker and target systems could communicate through the private network.
- Ensured that the vulnerable VM was not directly exposed to an external or production network.
- Used the Kali Linux address 192.168.56.102 as the local host address for the testing environment.

4. IP Address Configuration

- Opened the terminal in Kali Linux and executed: ifconfig
- Verified the Kali Linux IP address as 192.168.56.102.
- Confirmed the target Metasploitable2 address as 192.168.56.103.
- Verified that both systems belonged to the same 192.168.56.0/24 subnet.
- Configured the target address in Metasploit using: set RHOSTS 192.168.56.103
- Configured the Kali Linux address using: set LHOST 192.168.56.102

5. Network Connectivity Verification

- Verified communication between Kali Linux and the vulnerable VM over the isolated network.
- Performed an aggressive Nmap scan using: nmap -A 192.168.56.103
- Used Nikto to scan the target's HTTP service using: nikto -h http://192.168.56.103
- Reviewed the scan results and identified vsftpd 2.3.4 on port 21 as a candidate vulnerable service.
- Launched the Metasploit Framework using: msfconsole
- Searched for the relevant module using: search vsftpd
- Selected the identified vsftpd_234_backdoor module.
- Configured the target and local addresses using RHOSTS and LHOST.
- Used: show options to verify the module configuration.
- Attempted the controlled test against the vulnerable VM and recorded the observed outcome.

OBSERVATIONS:

During the cybersecurity practical, Kali Linux and Metasploitable2 were deployed on an isolated 192.168.56.0/24 Host-Only network. Nmap and Nikto were used to analyze the target, followed by controlled vulnerability testing using Metasploit.

- The Nmap scan successfully identified multiple services including vsftpd 2.3.4, OpenSSH, Telnet, SMTP, Apache, MySQL, PostgreSQL, VNC, and IRC, along with an outdated Linux kernel.
- The Nikto scan identified an outdated Apache/PHP installation, missing security headers, directory indexing, exposed phpinfo.php, and an accessible phpMyAdmin change log.
- The vsftpd 2.3.4 service was identified as a known vulnerable version, and the corresponding vsftpd backdoor exploit module was found in Metasploit.
- The Metasploit module was configured with RHOSTS 192.168.56.103 and LHOST 192.168.56.102 after identifying the Kali Linux IP address.
- The initial exploitation attempt failed because LHOST was not configured, but the error was resolved and the module was later fully configured; a successful callback or session was not confirmed in the provided output.

OUTPUT:

Phase 1 – Kali Linux Deployment

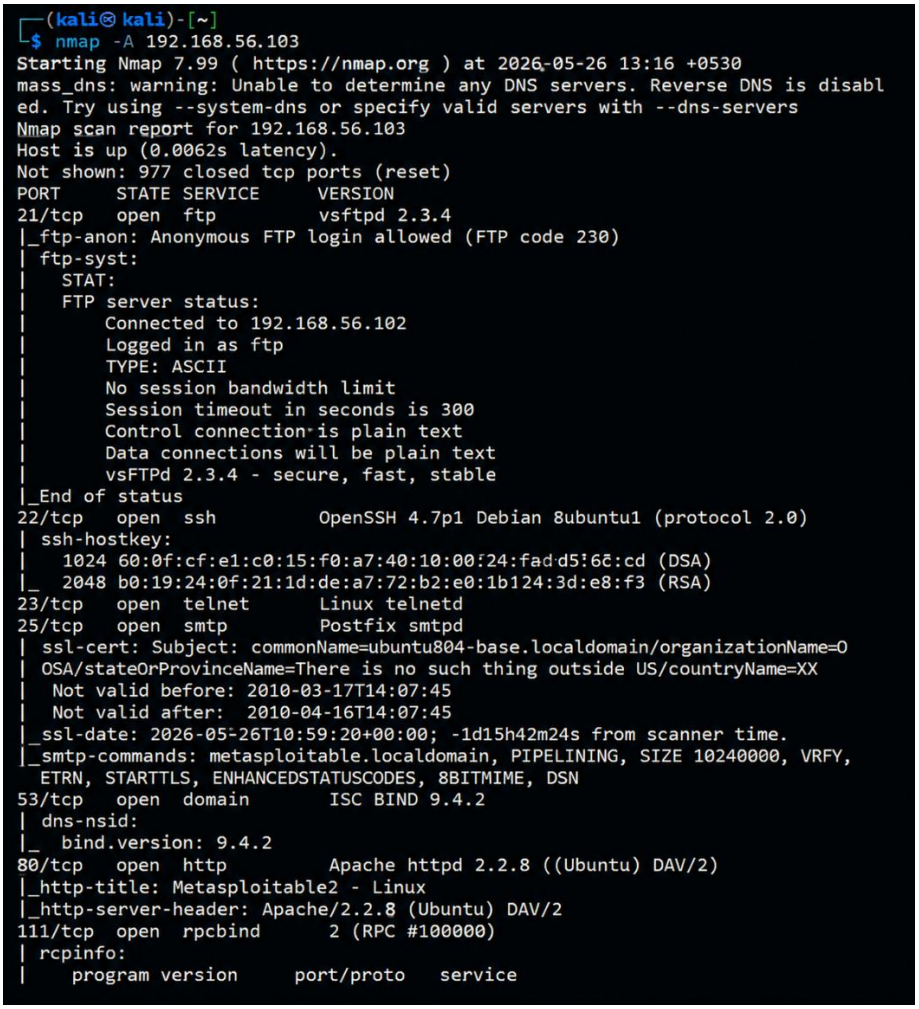


Fig. 1: : Nmap aggressive scan (-A) initiated against the target host 192.168.56.103

Phase 2 – Vulnerable VM Deployment

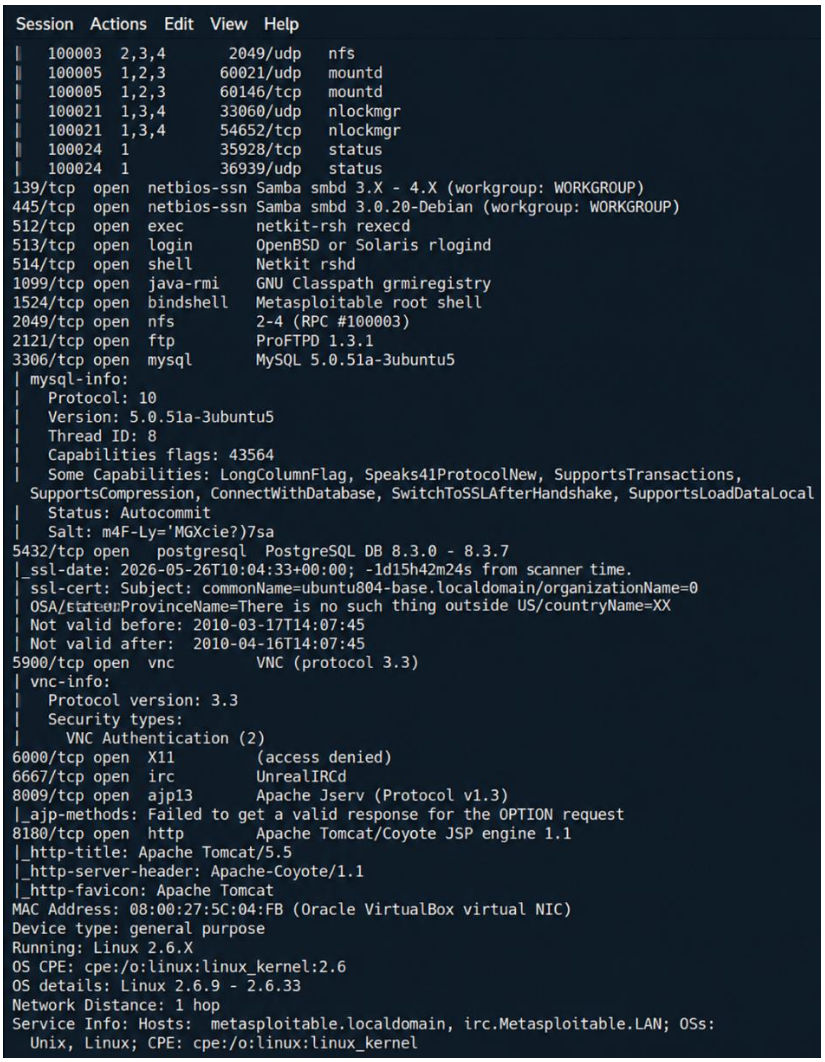


Fig. 2: Nmap Scan Results: RPC, NFS, SMB, MySQL, PostgreSQL, VNC, and IRC enumeration

Phase 3 – Isolated Virtual Network Configuration

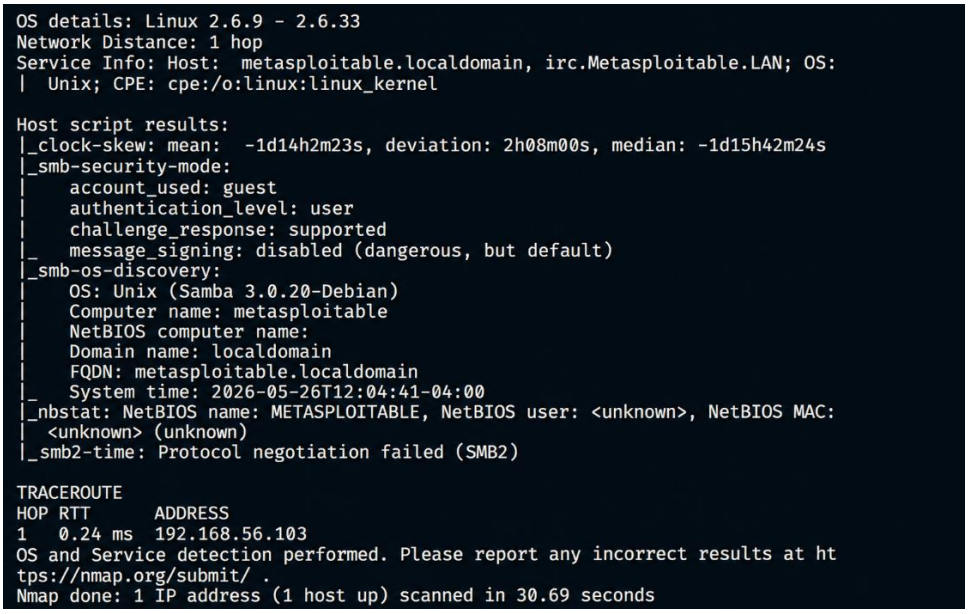


Fig. 3: Nmap OS detection, SMB host script results, and traceroute to the target

Phase 4 – IP Address Configuration

```
$ nikto -h http://192.168.56.103
-Nikto v2.0.0
---------------------------------------------------------------------------
Target IP:          192.168.56.103
Target Hostname:    192.168.56.103
Target Port:        80
Start Time:         2026-05-26 12:20:29 (GMT5.5)
---------------------------------------------------------------------------
Server: Apache/2.2.8 (Ubuntu) DAV/2
[ERROR] Failed to check for updates.
[INFO]   /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
[73939] /icons/: Directory indexing found.
[0] CGIs found: /cgi-bin/ .  Use '-C all' to force check all possible dirs). CGI tests skipped.
[65498]  /index: Uncommon header(s) 'tcn' found, with contents: list.
[62725]  /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force
file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=469ebdc59d4
[60805]  /PHP/5.2.4-2ubuntu5.10 appears to be outdated (current is at least 8.5.1).
[60903]  Apache/2.2 appears to be outdated (current is at least 2.4.60).
[81237]  /: Suggests a security header containing sensitive information. See: https://developer.mozilla.org/en-US/docs/
eb/HTTP/Headers/X-Content-Type-Options
[81237]  /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/
HTTP/CSP
[81237]  /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/He
aders/Referrer-Policy
[81232]  /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/
Web/HTTP/Headers/Strict-Transport-Security
[81233]  /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTT
P/Headers/Permissions-Policy
[99942]  /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
[00048]  /: HTTP TRACE method is active and possible which suggests the host is vulnerable to XST. See: https://owasp./
og/ww-community/attacks/Cross_Site_Tracing
[32056]  /phpinfo.php: Output from the phpinfo() function was found.
[75088]  /doc/: Directory indexing found.
[18103]  /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: https://cwe.mitre.org/cgi-bin/cvename.cgi?name=CWE-548
[68134]  /PMAREK0SF2A0-2C32-11d2-A3A0-AC2706C1A880/: PHP Easter Egg reveals potentially sensitive information via specific
HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/rd?tag=cve-2007-4887
[77171]  /php-easter-egg/:
[68134]  /PMAREK0SF2A0-2C32-11d2-A3A0-AC2706C1A880/: PHP Easter Egg reveals potentially sensitive information via specific
HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/rd?tag=cve-2007-4887
[77171]  /php-easter-egg/:
[68135]  /pma/:
[68134]  /PMAREK0SF2A0-2C32-11d2-A3A0-AC2706C1A880/: PHP Easter Egg reveals potentially sensitive information via specific
HTTP requests that contain specific QUERY strings. See: https://labs.detectify.com/rd?tag=cve-2007-4887
[77171]  /php-easter-egg/:
[88115]  /phpMyAdmin/ChangeLog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized
hosts.
[92033]  /phpMyAdmin/ChangeLog: Server may leak inodes via ETags, header found with file  /phpMyAdmin/ChangeLog. Inode:
92449, size: 45460, mtime: Tue Dec  9 23:54:00 2008. See: https://cwe.mitre.org/cgi-bin/cvename.cgi?name=CWE-158
---------------------------------------------------------------------------
+ 7993 requests: 0 error(s) and 18 item(s) reported on remote host
+ End Time:         2026-05-26 12:21:02 (GMT5.5) (33 seconds)
---------------------------------------------------------------------------
```

Fig. 4: Nikto web vulnerability scan against the target's HTTP service (port 80)

Phase 5 – Network Connectivity Verification

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Start commands with a space to avoid saving them to history

       =[ metasploit v6.4.133-dev                               ]
+ -- --=[ 2,647 exploits - 1,335 auxiliary - 2,141 payloads     ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
```

Fig. 5: Metasploit Framework console (msfconsole) launched on Kali Linux

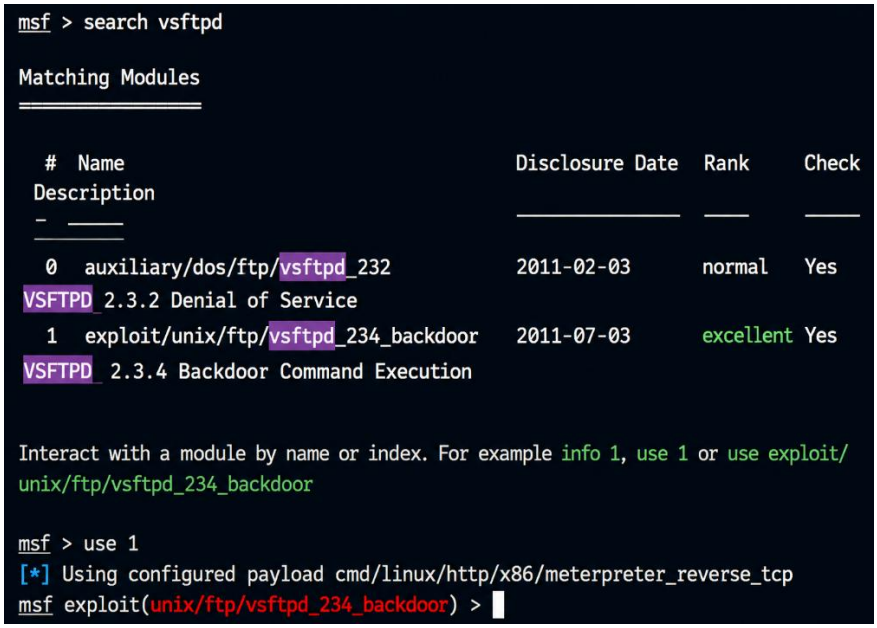


Fig. 6: Searching the module database for 'vsftpd' and selecting the backdoor exploit module

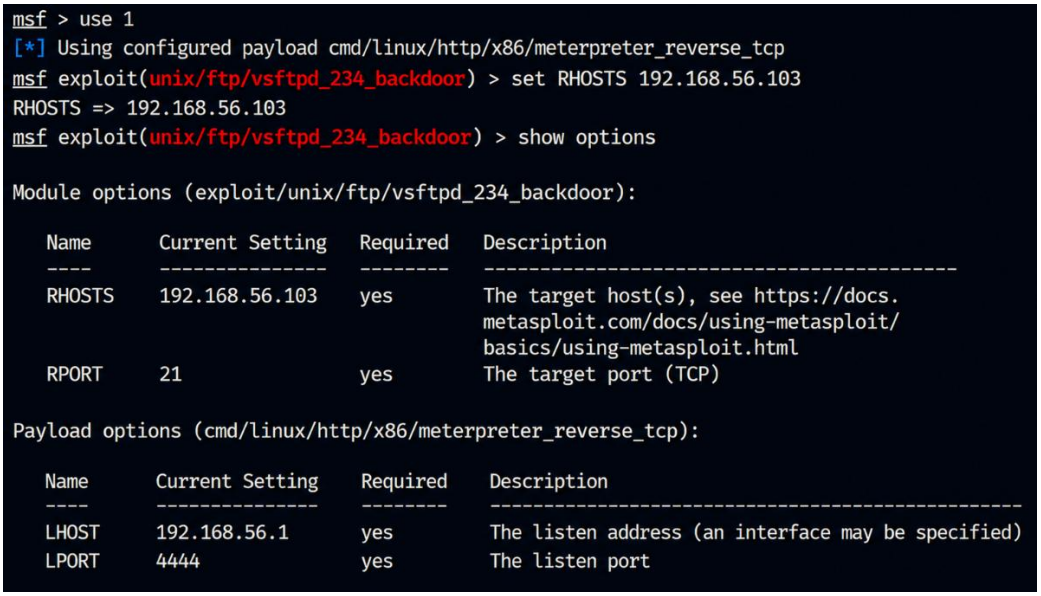


Fig. 7: Nmap OS detection, SMB information, and traceroute results

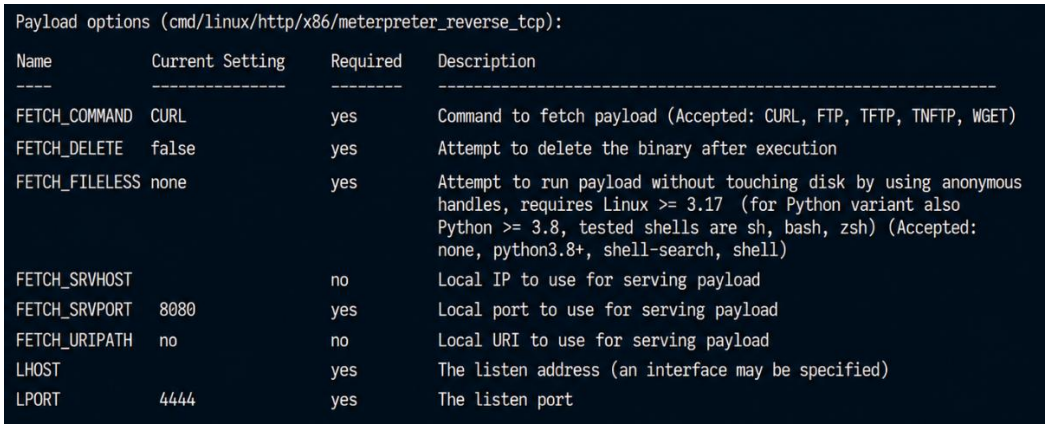


Fig. 8: Payload delivery options for the selected module, including FETCH_COMMAND and FETCH_SRVPORT

```
When FETCH_COMMAND is one of CURL,GET,WGET:

Name          Current Setting    Required    Description
----          ---------------    --------    -----------
FETCH_PIPE    false              yes         Host both the binary payload and
                                             the command so it can be piped di
                                             rectly to the shell.

When FETCH_FILELESS is none:

Name              Current Setting    Required    Description
----              ---------------    --------    -----------
FETCH_FILENAME    QdsNQvgkkc         no          Name to use on remote syste
                                                 m when storing payload; can
                                                 not contain spaces or slash
                                                 es
FETCH_WRITABLE_D  ./                 yes         Remote writable dir to stor
IR                                               e payload; cannot contain s
                                                 paces
```

Fig. 9: Additional payload options, including FETCH_PIPE and FETCH_FILENAME

```
Exploit target:

   Id   Name
   --   ----
   0    Linux/Unix Command


View the full module info with the info, or info -d command.
```

Fig. 10: Exploit target listing confirming the 'Linux/Unix Command' target

```
View the full module info with the info, or info -d command.

msf6 > info exploit/multi/script/web_delivery
[*] Displaying exploit/multi/script/web_delivery information (use info -d for extra details)

msf6 > info -d exploit/multi/script/web_delivery
[*] Displaying exploit/multi/script/web_delivery documentation
```

Fig. 11: Initial exploitation attempt failing due to an unset LHOST parameter

```
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
       inet 192.168.56.102  netmask 255.255.255.0  broadcast 192.168.56.255
       inet6 fe80::a00:27ff:fe4c:2a07  prefixlen 64  scopeid 0x20<link>
       ether 08:00:27:c4:2a:07  txqueuelen 1000  (Ethernet)
       RX packets 45  bytes 1500 (1.5 KiB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 65  bytes 8792 (8.5 KiB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
       inet 127.0.0.1  netmask 255.0.0.0
       inet6 ::1  prefixlen 128  scopeid 0x10<host>
       loop  txqueuelen 1000  (Local Loopback)
       RX packets 10232  bytes 831540 (812.0 KiB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 10252  bytes 831540 (812.0 KiB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Fig. 12: Identifying the Kali attacker machine's own IP address using ifconfig

```
┌──(kali㉿kali)-[~]
└─$ set LHOST 192.168.56.102

┌──(kali㉿kali)-[~]
└─$ show options
Command 'show' not found, but can be installed with:
sudo apt install nmh
sudo apt install mailutils-mh
```

Fig. 13: Attempt to set LHOST, followed by a command-context error

```
Global Options:
===============

   Option              Current Setting   Description
   ------              ---------------   -----------
   ConsoleLogging      false             Log all console input and output
   LogLevel            0                 Verbosity of logs (default 0, max 5)
   MeterpreterPrompt                     The meterpreter prompt string
   MinimumRank         manual            The minimum rank of exploits that wi
                                         ll run without explicit confirmation
                                         (Accepted: manual, low, average, no
                                         rmal, good, great, excellent)

   Prompt                                The prompt string
   PromptChar                            The prompt character
   PromptTimeFormat                      Format for timestamp escapes in prom
                                         pts

   SessionLogging      false             Log all input and output for session
                                         s
   SessionTlvLogging                     Log all incoming and outgoing TLV pa
                                         ckets
   TimestampOutput     false             Prefix all console output with times
                                         tamp
   VERBOSE             false             Enable detailed status messages wher
                                         e specific behavior can differ per m
                                         odule

msf > exploit
[-] Unknown command: exploit. Run the help command for more details.
```

Fig. 14: Reviewing global console options while resolving the module context, followed by an unrecognised 'exploit' command

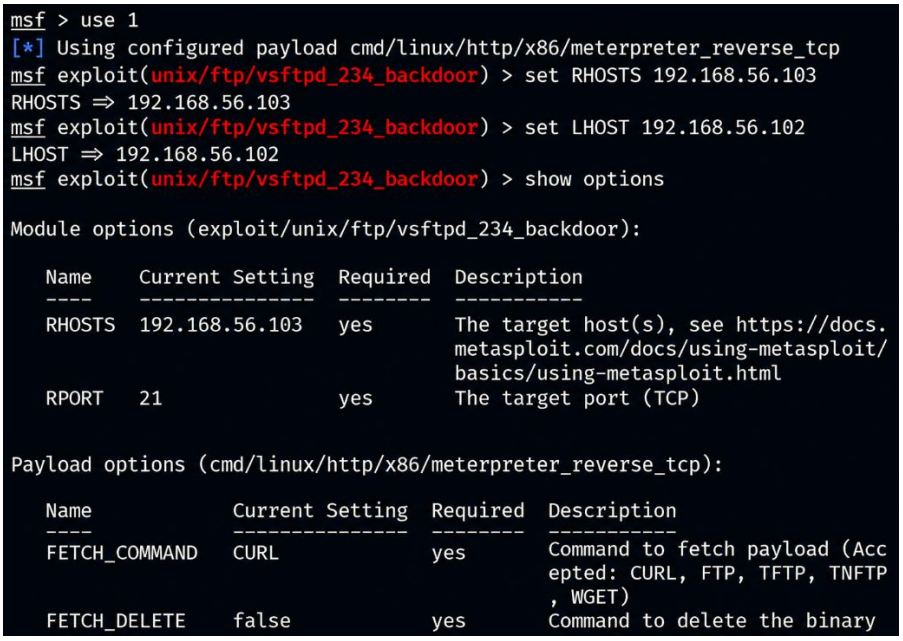


Fig. 15: Final module configuration with RHOSTS and LHOST both correctly set

ANALYSIS:

The experiment successfully demonstrated the deployment of Kali Linux and Metasploitable2 on an isolated network. Nmap and Nikto identified open services and vulnerabilities, while Metasploit was used to configure and test the identified vsftpd 2.3.4 vulnerability. The final configuration was completed, but successful exploitation was not confirmed.

PREVENTIVE MEASURES:

1. Keep operating systems, applications, and network services regularly updated with security patches.
2. Disable unnecessary services and close unused ports to reduce the attack surface.
3. Use strong authentication and secure protocols such as SSH, SFTP, and HTTPS.
4. Isolate vulnerable systems from production networks using firewalls and dedicated virtual networks.
5. Perform regular vulnerability scanning and security audits to identify and fix weaknesses.

CONCLUSION:

The experiment successfully demonstrated the deployment and network configuration of Kali Linux and Metasploitable2 on an isolated virtual network. Network scanning and vulnerability assessment were performed using Nmap, Nikto, and Metasploit, demonstrating the importance of network isolation and regular security testing.